

Yadnesh Krishnakant Bhanushali

Applied AI / ML Engineer | AWS Architecture | Generative AI | Search Systems
Dombivli, Maharashtra 📞 +91-7021438481 ✉ yadneshkb1303@gmail.com
🌐 linkedin.com/in/yadnesh-bhanushali-554678265 🐙 github.com/yadnesh13

Professional Summary

Applied AI / Machine Learning Engineer experienced in designing and deploying production AI systems across computer vision, generative AI, hybrid search, and cloud-based industrial platforms. Skilled at translating ambiguous business requirements into scalable architectures and delivering end-to-end solutions spanning models, data pipelines, cloud infrastructure, and monitoring systems. Known for strong ownership, rapid execution, and independent delivery of enterprise AI solutions in real client environments.

Technical Skills

Cloud & Architecture: AWS (SageMaker, Bedrock, RDS, VPC, IAM, S3), Cloud Architecture Design, Infrastructure Security
AI / ML: LLMs, VLMs, Computer Vision, Generative AI, Real-ESRGAN, Learning-to-Rank (LTR), Hybrid Search, Data Modelling, Data Pre-Processing
Search & Data Systems: OpenSearch, BM25, Semantic kNN Search, Re-ranking Pipelines, Data Engineering
Programming: Python, Java, SQL, C++, PostgreSQL, HTML, CSS **Tools:** Grafana, Streamlit, Django, Git

Experience

Oneture Technologies

Oct 2025 – Present

Associate Machine Learning Engineer (Converted from ML Intern)

Mumbai, India

- **Hybrid AI Search Engine (PoC):** Sole developer of a high-throughput OpenSearch hybrid retrieval system on AWS, combining BM25 lexical search with semantic kNN embeddings, custom neural re-ranking, and Learning-to-Rank personalization — designed to handle **large-scale retrieval workloads** across multiple e-commerce user personas.
- **Multimodal AI Pipeline — Snapdeal & Nykaa (PoC → Pilot → Live):** Built and took to production a Vision Language Model system via AWS Bedrock that extracts structured product attributes, banner elements, semantic callouts, layout compliance scores, and visual metadata from large-scale e-commerce imagery. System is fully config-driven — operators control target detection zones, supported aspect ratios, and extraction rules from a single config file. Outputs downloadable Excel reports with extracted attributes, compliance flags, and layout scores. **Significantly reduced manual cataloguing effort by 80 %**, with zero human intervention required in the live pipeline after a 3-month pilot.
- **Generative Vision Pipelines:** Built SageMaker ML pipelines for image super-resolution and outpainting using Real-ESRGAN, enabling automated marketing image transformation at scale; deployed with a Streamlit GUI for non-technical stakeholder use.
- **Digital Twin Platform — Wind Energy (Live):** Contributed to AWS-based architecture for an industrial wind turbine Digital Twin platform covering telemetry ingestion, condition monitoring, and predictive analytics readiness. Developed multi-level Grafana dashboards (site, turbine, component) used by real site operators in production.
- **TEAL Demand Forecasting Platform:** Sole owner of database design and deployment for a client demand forecasting system. Planned, designed, and deployed AWS RDS per client specifications across secure VPC/subnet configurations. Built three ingestion pipelines: (1) bulk upload for initial data load, (2) record update pipeline for corrections and amendments, and (3) an incremental upsert pipeline for monthly data — intelligently updating quantity for existing inventory and sales records rather than overwriting, and inserting only new entries, preventing data duplication and preserving historical integrity.
- **Mentorship:** Guided two junior interns on ML implementation tasks alongside delivery of production projects.

Projects

AyurGenix AI 2024–2025 🐙 GitHub — AI-powered Ayurvedic advisory system using LLM reasoning and ensemble ML models. Full-stack pipeline with Django and PostgreSQL; achieved **88% treatment mapping accuracy**. Published at ICETCSE-2025.

PDF MasterMind 2024 🐙 GitHub — LLM-powered document assistant for conversational querying and summarization of large PDFs using LangChain and transformer-based retrieval pipelines.

Education

Datta Meghe College of Engineering

B.E. Artificial Intelligence & Data Science (Honors in Machine Learning)

2021 – 2025

CGPA: 9.04

K.V. Pendharkar College of Science

Higher Secondary Certification (PCMB)

2019 – 2021

85%

Achievements & Publications

Research Publication: Presented "AyurGenix AI — Ayurvedic Medication Advisor" at the **International Conference on Emerging Trends in Computer Science and Engineering (ICETCSE-2025)**, Navi Mumbai, India.

Hackathon: Optimized algorithmic pipeline at CodeRush @ LogiTHON, improving solution efficiency by **20%**.